

$$\begin{aligned}
 1) \ z_1^{(1)} &= w_{11}x_1 + w_{12}x_2 + b_1^{(1)} \\
 &= (0.5)(1) + (-0.3)(2) + 0.1 \\
 &= \boxed{0}
 \end{aligned}$$

$$\begin{aligned}
 2) \ h_1 &= \text{RELU}(z_1^{(1)}) \\
 &= \boxed{0}
 \end{aligned}$$

$$\begin{aligned}
 3) \ z_2^{(1)} &= w_{21}x_1 + w_{22}x_2 + b_2^{(1)} \\
 &= (0.8)(1) + (0.2)(2.0) + (-0.1) \\
 &= \boxed{1.1}
 \end{aligned}$$

$$4) \ h_2 = \text{ReLU}(z_2^{(1)}) = \boxed{1.1}$$

$$\begin{aligned}
 5) \ z^2 &= v_1h_1 + v_2h_2 + b^{(2)} = (0.6)(0) + (-0.4)(1.1) + (0.2) \\
 &= \boxed{-0.24}
 \end{aligned}$$

$$6) \ \hat{y} = \sigma(z^{(2)}) =$$

$$\sigma = \frac{1}{1 + e^{-x}} = \boxed{0.44028}$$

$$\begin{aligned}
 \Rightarrow L &= -[y \log(\hat{y}) + (1-y) \log(1-\hat{y})] \\
 &= \boxed{-0.8203} = \boxed{0.82032}
 \end{aligned}$$